Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC studying parallel lines and transversals, $m\angle A = 53^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 75°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (53^\circ + 52^\circ) = 75^\circ$.

2 **[Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In ΔABC studying triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 81°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (54^\circ + 45^\circ) = 81^\circ$.

3 **[Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after t hours, and find the population after 12 hours.

Initial population $a = 250$, doubling factor $b = 2$, time interval = 3 hours: $B(t) = 250 \cdot 2^{t/3}$. For $t = 12$ hours, the number of doubling periods is $12/3 = 4$: $B(12) = 250 \cdot 2^4 = 250 \cdot 16 = 4000$ bacteria.

4 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Find all solutions to $2x^2 - 6x + 5 = 0$ using the quadratic formula, and determine whether the roots are real or complex.

Identify coefficients: $a = 2$, $b = -6$, $c = 5$. Evaluate the discriminant:
$\Delta = b^2 - 4ac = (-6)^2 - 4(2)(5) = 36 - 40 = -4$. Because \(\Delta

5 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

6 **[Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Quadratic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

[Functions: Composite & Inverse · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

[Functions, Modulus & Quadratic Theory · Practice 1] If the governing relation for this system is $g(t) = 3t + 3$, evaluate $g(4)$ and find the inverse input t when $g(t) = 18$.

Substitute $t = 4$: $g(4) = 3(4) + 3 = \mathbf{15}$. Setting $3t + 3 = 18$ gives $3t = 15 \Rightarrow t = \mathbf{5}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A solid metal sphere of radius 3 cm is melted down and recast into a cylinder of radius 2 cm. What is the height of the recast cylinder in cm?

Step 1: Calculate the volume of the sphere: $V_{\text{sphere}} = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3^3) = \frac{4}{3}\pi(27) = 36\pi$ cm^3. Step 2: Equate with the volume of the cylinder: $V_{\text{cyl}} = \pi R^2 h = \pi(2^2)h = 4\pi h$. Step 3: Solve for height h: $4\pi h = 36\pi \Rightarrow h = \frac{36\pi}{4\pi} = 9$ cm. The height of the cylinder is 9 cm.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

(a) Use $u_n = a + (n-1)d$: $u_3 = a + 2d = 14$ $u_7 = a + 6d = 34$ Subtract the first equation from the second: $4d = 20 \Rightarrow d = 5$. Substitute back: $a + 2(5) = 14 \Rightarrow a = 4$. (b) Use the sum formula $S_n = \frac{n}{2}(2a + (n-1)d)$ with $n = 25$: $S_{25} = \frac{25}{2}(2(4) + 24(5)) = \frac{25}{2}(8 + 120) = \frac{25}{2}(128) = 25 \times 64 = 1600$

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

(a) $F_f = 0.4 \times 3 \times 9.8 = 11.76$ N. (b) $a = \frac{20 - 11.76}{3} \approx 2.75$ m s^{-2}. (c) $t = \sqrt{10/2.75} \approx 1.91$ s. Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Factor the quadratic expression $6x^2 + 11x - 10$ completely.

Use the ac-method. We seek two integers whose product is $a \cdot c = 6(-10) = -60$ and whose sum is $b = 11$. The factor pair of -60 that sums to 11 is 15 and -4 (since $15 \times (-4) = -60$ and $15 + (-4) = 11$). Rewrite the middle term and factor by grouping: $6x^2 + 15x - 4x - 10 = 3x(2x + 5) - 2(2x + 5)$. Factor out the binomial $(2x + 5)$: $(2x + 5)(3x - 2)$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A circle has radius 8 cm. A sector of this circle has a total perimeter of 28 cm. Find the angle of the sector in radians, and calculate the exact area of the sector.

The perimeter of a sector is given by $P = s + 2r = r\theta + 2r$. Substitute $P = 28$ and $r = 8$: $28 = 8\theta + 2(8) \Rightarrow 28 = 8\theta + 16 \Rightarrow 8\theta = 12 \Rightarrow \theta = \frac{12}{8} = 1.5$ radians The area of the sector is: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)(1.5) = \frac{1}{2}(64)(1.5) = 32 \times 1.5 = 48$ cm^2

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

Find the critical values where $2x^2 - 7x + 3 = 0$. Factor the quadratic: $(2x - 1)(x - 3) = 0$, so the roots are $x = \frac{1}{2}$ and $x = 3$. Since the coefficient of x^2 is positive ($a = 2 > 0$), the parabola opens upward. The quadratic is less than or equal to zero between and including the roots. Therefore, the solution set is $\frac{1}{2} \leq x \leq 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Geometry

Simplify the algebraic expression $\frac{(2x^3y^{-2})^3}{(4x^{-1}y^4)^2}$ leaving only positive exponents.

Apply power of a power to numerator and denominator separately: Numerator $= 2^3 \cdot (x^3)^3 \cdot (y^{-2})^3 = 8x^9y^{-6}$. Denominator $= 4^2 \cdot (x^{-1})^2 \cdot (y^4)^2 = 16x^{-2}y^8$. Divide coefficients and subtract exponents: $\frac{8}{16} \cdot x^{9-(-2)} \cdot y^{-6-8} = \frac{1}{2}x^{11}y^{-14} = \frac{x^{11}}{2y^{14}}$.

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16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Trigonometry**

Two parallel lines L_1 and L_2 are cut by a transversal T. A pair of alternate exterior angles are given by $(7x - 14)^\circ$ and $(4x + 31)^\circ$. Find the value of x and the measure of each alternate exterior angle.

Step 1: Set up the equation using alternate exterior angles: Alternate exterior angles created by parallel lines are congruent: $7x - 14 = 4x + 31 \Rightarrow 3x = 45 \Rightarrow x = 15$. Step 2: Evaluate the angle measure: $7(15) - 14 = 105 - 14 = 91^\circ$. Check: $4(15) + 31 = 60 + 31 = 91^\circ$. Both alternate exterior angles measure 91°.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Transformations & Vectors**

Each interior angle of a regular polygon measures 144°. How many sides does the polygon have?

Solution: The fastest approach is to find the exterior angle first. Since interior and exterior angles form a linear pair: $\text{Exterior angle} = 180^\circ - 144^\circ = 36^\circ$. Since the exterior angles of any regular n-gon sum to 360°: $n = \frac{360^\circ}{36^\circ} = 10$. The polygon is a regular decagon with 10 sides.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics**

Two angles $\angle P$ and $\angle Q$ form a linear pair. If $m\angle P = (4x + 12)^\circ$ and $m\angle Q = (2x + 18)^\circ$, find the measure of the smaller angle.

Solution: Since they form a linear pair, their measures sum to 180°: $(4x + 12) + (2x + 18) = 180 \Rightarrow 6x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$. Now substitute $x = 25$ back into each expression: $m\angle P = 4(25) + 12 = 112^\circ$, $m\angle Q = 2(25) + 18 = 68^\circ$. The smaller angle has measure 68°.

19 **[Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number**

Simplify and rationalize the denominator: $\frac{\sqrt{3}}{\sqrt{5} - \sqrt{2}}$.

Multiply numerator and denominator by the conjugate $\sqrt{5} + \sqrt{2}$: $\frac{\sqrt{3}(\sqrt{5} + \sqrt{2})}{(\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})}$. Expand denominator using difference of squares $(a - b)(a + b) = a^2 - b^2$: $(\sqrt{5})^2 - (\sqrt{2})^2 = 5 - 2 = 3$. Expand numerator and simplify: $\frac{\sqrt{15} + \sqrt{6}}{3}$.

20 **[Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Sets & Venn Diagrams**

In right ΔABC with right angle at C, altitude CD is drawn to hypotenuse AB. If $AD = 4$ and $DB = 9$, find the length of leg BC.

Step 1: Compute total hypotenuse AB: $AB = AD + DB = 4 + 9 = 13$ Step 2: Apply the Geometric Mean Leg Rule: The leg BC is the geometric mean of the hypotenuse AB and the adjacent segment DB: $BC^2 = AB \cdot DB = 13 \cdot 9 = 117$ Step 3: Simplify the radical: $BC = \sqrt{117} = \sqrt{9 \times 13} = 3\sqrt{13} \approx 10.82$

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying geometrical diagrams, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 84° ✓

b. 94°

c. 76°

d. 129°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (51^\circ + 45^\circ) = 84^\circ$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In ΔABC studying measuring angles, $m\angle A = 52^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 83° ✓

b. 93°

c. 75°

d. 128°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (52^\circ + 45^\circ) = 83^\circ$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 2x + 8$ in quadrilaterals, evaluate $f(4)$.

a. 16 ✓

b. 18

c. 8

d. 20

Substitute $x = 4$: $f(4) = 2(4) + 8 = 16$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the value of x that must be excluded from the domain of $f(x) = \frac{1}{x-6}$.

a. 11

b. **6** ✓

c. 1

d. 15

The denominator cannot be zero: $x - 6 \neq 0 \Rightarrow x \neq 6$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the smallest integer x for which $f(x) = \sqrt{x - 6}$ is defined.

a. **6** ✓

b. 0

c. 13

d. -3

Need $x - 6 \geq 0 \Rightarrow x \geq 6$. Smallest integer: 6.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Given that $x = 3.6 \times 10^5$ and $y = 4.8 \times 10^4$, find the value of $x + y$ in standard form.

a. 4.08×10^5 ✓

b. 8.4×10^5

c. 4.08×10^9

d. 3.648×10^5

Align exponents to 10^5: $y = 0.48 \times 10^5$. Then $x + y = (3.6 + 0.48) \times 10^5 = 4.08 \times 10^5$. [M1 for matching powers of 10, A1 for 4.08x10^5]

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Write 0.00045 in standard form.

a. 4.5×10^{-4} ✓

b. 4.5×10^{-5}

c. 45×10^{-5}

d. 4.5×10^{-3}

Move the decimal point 4 places to the right to obtain 4.5. Since the original number is less than 1, the exponent is negative: 4.5×10^{-4}. [B1]

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The length $L = 12$ cm and width $W = 8$ cm of a rectangle are each measured correct to the nearest centimeter. What is the UPPER bound for the area?

a. **106.25 cm²** ✓

b. 96 cm²

c. 117.00 cm²

d. 104.25 cm²

Upper bound of length = $12 + 0.5 = 12.5$. Upper bound of width = $8 + 0.5 = 8.5$. Max area = $12.5 \times 8.5 = 106.25$ cm^2. [M1 A1]

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The ratio of boys to girls in a school is (4 : 5). There are 180 girls. How many students are there in total in the school?

a. 324 ✓

b. 144

c. 225

d. 360

One ratio part equals $\frac{180}{5} = 36$, and the total number of parts is $4 + 5 = 9$, so the total number of students is $9 \times 36 = 324$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A car travels a distance of 140 km, correct to the nearest 10 km, in a time of 2.0 hours, correct to the nearest 0.1 hour. What is the LOWER bound of the average speed?

a. 65.85 km/h ✓

b. 70.00 km/h

c. 74.36 km/h

d. 67.50 km/h

Lower bound of distance = 135 km. Upper bound of time = 2.05 h. Lower bound of speed = $\frac{135}{2.05} \approx 65.85$ km/h. [M1 A1]

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A jacket is sold for \68 after a 15% discount. What was the original price of the jacket?

a. *x*78.20

b. *x***80.00 ✓**

c. *x*82.50

d. *x*85.00

The sale price represents $100\% - 15\% = 85\%$ of the original price. Original price = *x*80. [M1 for dividing by 0.85, A1 for 80]

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A length is measured as 14 cm correct to the nearest centimetre. What is the lower bound of the length?

a. 13 cm

b. 13.4 cm

c. **13.5 cm** ✓

d. 13.9 cm

Since the degree of accuracy is to the nearest 1 cm, the error is ± 0.5 cm. Lower bound = $14 - 0.5 = 13.5$ cm. [B1]

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two similar geometric solids have a linear scale factor of 1 : 2. If the smaller solid has volume 50 cm³, what is the volume of the larger solid?

a. **400 cm³** ✓

b. 200 cm³

c. 100 cm³

d. 500 cm³

For similar solids, volume scale factor = $k^3 = 2^3 = 8$. Larger volume = $50 \times 8 = 400$ cm³. [M1 A1]

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Convert 8000 cm³ into litres.

a. **8 litres** ✓

b. 80 litres

c. 0.8 litres

d. 800 litres

1 litre $= 1000$ cm³. 8000 cm³ $= \frac{8000}{1000} = 8$ litres. [B1]

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Convert 35% to a fraction in lowest terms.

a. $\frac{35}{100}$

b. $\frac{7}{20}$ ✓

c. $\frac{7}{10}$

d. $\frac{3}{5}$

Applying Number principles, $\frac{35}{100} = \frac{7}{20}$ Therefore, the verified correct option is $\frac{7}{20}$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A circle has an area of 49π cm^2. What is its circumference in terms of π?

a. 14π cm ✓

b. 7π cm

c. 28π cm

d. 21π cm

Area = $\pi r^2 = 49\pi \Rightarrow r = 7$ cm. Circumference = $2\pi r = 14\pi$ cm. [M1 A1]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A solid hemisphere has volume 18π cm^3. What is its radius?

a. **3 cm** ✓

b. 6 cm

c. 9 cm

d. 2 cm

Volume of hemisphere = $\frac{2}{3}\pi r^3 = 18\pi \Rightarrow r^3 = 27 \Rightarrow r = 3$ cm. [M1 A1]

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics

Divide 420 in the ratio 2 : 3 : 5. What is the value of the largest share?

a. **210** ✓

b. 126

c. 84

d. 250

Total parts = $2 + 3 + 5 = 10$. 1 part = $\frac{420}{10}$ = \)42. Largest share = $5 \times 42 =$ 210. [M1 A1]

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number

A solid cylinder has radius 4 cm and height 10 cm. Find its total surface area in terms of π.

a. 112π cm^2 ✓

b. 80π cm^2

c. 96π cm^2

d. 160π cm^2

Total SA = $2\pi rh + 2\pi r^2 = 2\pi(4)(10) + 2\pi(16) = 80\pi + 32\pi = 112\pi$ cm^2. [M1 A1]

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Sets & Venn Diagrams

Calculate the total value of an investment of 2000 after 3 years at an interest rate of 4% per year compound interest.

a. **2249.73** ✓

b. 2240.00

c. 2299.73

d. 2080.00

Formula: $A = P(1 + \frac{r}{100})^n = 2000(1 + \frac{4}{100})^3$ = 2249.73. [M1 A1]

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Algebra & Graphs

Calculate the volume of a sphere with radius 3 cm in terms of π.

a. 36π cm^3 ✓

b. 12π cm^3

c. 108π cm^3

d. 27π cm^3

Volume = $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi(3)^3 = \frac{4}{3}\pi(27) = 36\pi$ cm^3. [M1 A1]

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Coordinate Geometry

A cone has base diameter 12 cm and vertical height 8 cm. What is its volume in terms of π?

a. 96π cm^3 ✓

b. 288π cm^3

c. 48π cm^3

d. 192π cm^3

Radius = 6 cm. Volume = $\frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(8) = 96\pi$ cm^3. [M1 A1]

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Mensuration

A trapezium has parallel sides of length 7 cm and 11 cm, and a perpendicular height of 6 cm. What is its area?

a. **54 cm²** ✓

b. 108 cm²

c. 42 cm²

d. 66 cm²

Area = $\frac{1}{2}(a + b)h = \frac{1}{2}(7 + 11)(6) = \frac{1}{2}(18)(6) = 54$ cm^2. [M1 A1]

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Geometry

Percentage increase from 40 to 50:

a. 10%

b. 20%

c. **25%** ✓

d. 50%

Applying Number principles, $\frac{10}{40} \times 100 = 25\%$ Therefore, the verified correct option is 25%.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

In a sale, all prices are reduced by 20%. The sale price of a jacket is 84. What was the ORIGINAL price of the jacket?

a. **105** ✓

b. 101

c. 115

d. 100

The sale price represents $100\% - 20\% = 80\%$ of the original price, so the original price is $x105$.

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